Promo Pulse: a new approach to Rossmann-style hub data, with four hypothesis tests on promotions, data gaps and competitors

Group name	DSC 10
Dataset	Shadow Hub database (training, test and hub files, 1,115 hubs): https://github.com/kushiiitd05/ShadowDsc10-ShadowHub/tree/cse558-midterm-2026/1_midterm/data/raw
Code	https://github.com/kushiiitd05/ShadowDsc10-ShadowHub/tree/main/1_midterm (scripts in 1_midterm/analysis/, README in 1_midterm/README.md)

#	Name	Roll no.	Institute email	Contribution
1	Manya Chaturvedi	2023312	manya23312@iiitd.ac.in	Chose the tests and the reasoning for each, with the alternatives we rejected
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1. Dataset

Source: the training, test and hub files of the Shadow Hub database, which our group received through a private data-science competition; the files are in our repository (link above). One row of orders_train.csv is one retail hub on one calendar day, 1 January 2013 to 19 June 2015 (900 dates, 1,115 hubs). hub_metadata.csv has one row per hub. orders_test.csv (20 June to 31 July 2015) has no outcome, so it is used only to describe the schedule.

Provenance: the files match the public Rossmann Store Sales data with renamed columns. Evidence: 1,115 hubs; $970{,}379 + 46{,}830 = 1{,}017{,}209$ rows, Rossmann's training size; hub 1 (format 3, tier 1, competitor distance 1,270, competitor since September 2008, no loyalty programme) and hub 2 (loyalty since week 13 of 2010, months "Jan,Apr,Jul,Oct") reproduce Rossmann stores 1 and 2; and the same 180 hubs are missing from 1 July to 31 December 2014.

	Raw file	After encoding
Instances (rows)	970,379	970,379
Attributes (columns)	9 (+ 9 hub fields)	46
Instances × attributes	8,733,411	**44,637,434**

Size check: $970{,}379 \times 46 = 44{,}637{,}434 \geq 3{,}000{,}000$; the raw file alone gives $970{,}379 \times 9 = 8{,}733{,}411$. The 46 attributes are: HubID; 4 calendar fields from Date; one-hot Weekday (7), RegionalHoliday (4), HubFormat (4) and loyalty interval (4, incl. "None"); 14 numeric or binary fields; and 8 data-quality flags (Section 2). Rows are counted as hub-days, but every test uses the *hub* as its independent unit (Section 5).

Key features

Feature	Type	Real-world meaning (verified facts)
OrderVolume	discrete	**response**: the hub's daily volume, *units unstated*. Median volume per app session is 9.23 on open days, which is implausible for a count of orders
AppSessions	discrete	app sessions (traffic); Pearson $r = 0.825$ with volume on open days
IsOpen	nominal	hub open that day (82.87% of rows)
PromoActive	nominal	promotion running (38.27%); set per date, identical for all hubs, in alternate Monday–Friday weeks, never at weekends
Weekday	ordinal (cyclic)	1 = Monday … 7 = Sunday; matches Date in every row
RegionalHoliday	nominal (4)	none, public holiday, Easter, Christmas
HubFormat	nominal (4)	store format; 602 / 17 / 148 / 348 hubs
CompetitorDistance	continuous	distance to the nearest competitor (median 2,325; 3 hubs missing)
CompetitorOpenSinceMonth, Year	discrete	month and year that competitor opened; unknown for 354 hubs

2. Data preparation

No raw value was overwritten; every step adds a flag or a new column, so each choice can be undone. "Why not" names the alternative we rejected.

Issue found	What we did	Why, and not the alternative
33,121 absent hub-dates: 180 hubs $\times$ 184 days (2014-H2) + hub 988 on 2013-01-01	left absent; flag `gap_hub`; tested in T2	a systematic block, not random gaps. Zero-filling would invent 33,121 closures; mean, median, kNN or regression imputation would invent 2014-H2 sales for exactly the hubs that T2 shows are not a random subset
166,269 closed rows, all with volume 0 and sessions 0	kept, excluded from volume tests	structural zeros, neither missing nor outliers; mixing "is it open?" with "how much?" gives a two-humped mixture
54 open rows with volume 0 (41 hubs)	flag `open_zero`; excluded; listed in README	an open day with no activity is inconsistent. MLE $\hat{p} = 54/804{,}110 = 6.7 \times 10^{-5}$ (Wilson 95% CI 5.1×10^{-5} to 8.8×10^{-5}), so dropping them cannot move any estimate
Pooled IQR fence (Q1 4,856, Q3 8,360, fence 13,616) flags 29,307 rows (3.64%)	not deleted; tests use $\log(1+V)$ and per-hub contrasts	88.3% of these rows belong to the 189 hubs whose *own* median exceeds Q3: the fence measures hub size, not error. Trimming or winsorising would cut big hubs' ordinary days
Per-hub robust z (median/MAD, $\lvert z\rvert > 3.5$): 6,220 rows, 1,040 hubs	kept as genuine demand	97.0% sit next to a closure (63.2%), before Christmas (17.5%), on or next to a holiday (11.1%), on promo (4.2%) or after a reopening (1.1%). The maximum, 38,722 (hub 262), is Good Friday 2015 with a promotion
Normal "3σ" rule	not used	$\lvert z\rvert \geq 3$ holds for 1.59% of open days against 0.27% under normality (kurtosis 7.84); the distribution-free Chebyshev bound (11.1%) holds, but a normal cut-off would mislabel real peaks
88 closure runs ≥ 5 days at 85 hubs (longest 185)	flags `long_closure`, 14-day `reopen_window`	refurbishment-like; volume in the 14 days after reopening is 44% above the 28 days before closing, so reopen days are excluded in a sensitivity rerun
Volume/session ratio: median 9.23, max 64.96 (hub 897: 16,954 / 261); 23 rows > 30	flag `ratio_anomaly`; kept	sessions are probably under-recorded; no test uses sessions
Metadata: distance missing (3 hubs); opening date missing (354); year 1900 (hub 815); 1961 (hub 146)	complete-case in T3 ($n = 1{,}112$); date left NA + indicator; 1900 $\rightarrow$ NA; 1961 kept	imputation would place hubs at invented distances or dates inside the very tests that use them; a 1900 opening means a 115-year-old competitor, a sentinel
Loyalty start/interval NA for 544 hubs	level "None"	exactly the 544 hubs without the programme: *not applicable*, not missing
Duplicates	none removed	0 duplicate hub-dates, 0 duplicate rows; 8 hubs form 4 pairs with identical metadata but different daily series, so they are distinct hubs
Integer codes for nominal fields	one-hot; tier as ordinal code; log scale as a new column	codes 1–4 for format would impose a false order and distance
Integrity	checked	last digit of volume (≥ 100) is uniform, $\chi^2(9) = 11.10$, $p = 0.27$: no heaping or rounding

3. What changed after preparation

Measure	Before	After
Rows / attributes	970,379 / 9	970,379 / 46 (no row deleted)
Rows that enter the tests	–	804,056 open rows with volume > 0
Missing cells after the hub join	2,075,025	1,591,274, of which 970,366 are *not applicable* loyalty fields
Skewness of open-day volume	1.59 (raw)	-0.11 on $\log(1+V)$
Kurtosis (normal $= 3$)	7.84 (raw)	3.65 on $\log(1+V)$
Flagged rows	–	open-zero 54; ratio 23; long closure 1,898; reopen window 1,218
Second outlier pass (robust z on $\log(1+V)$)	–	2,645 rows (2,060 low, 585 high); 1,870 explained by holiday, closure, reopening or pre-Christmas; the rest kept

Nothing was deleted: preparation turned silent problems into flags, and each test states which flags it excludes. The log scale removes most of the skew and the excess kurtosis, which is why every test uses it. The second outlier pass, run after the fixes as Lecture 3 asks, mostly flags *low* days next to closures and holidays, so it does not justify any further deletion.

4. Basic questions and problem statement



Figure 1: Open-day volume is right-skewed (skewness 1.59, kurtosis 7.84); on $\log(1+V)$ it is nearly symmetric (-0.11, 3.65), **so all tests use the log scale**.



Figure 2: **180 of 1,115 hubs (16.1%)** have no rows at all in 2014-H2; **format 4 loses 23.3%** of its hubs (Wilson 95% CIs; hubs per format in brackets).



Figure 3: The promotion lift **falls through the week, from 60% on Monday to 22% on Friday** (mean of 1,115 hub-level lifts; 95% CIs are shorter than the dots).



Figure 4: Hub volume is flat in competitor distance except in the closest decile (distance ≤ 250), whose 122 hubs **sell 15.9% more** than the rest.

Basic questions

1. Does a promotion raise volume? (Yes: by 22–60%, depending on the weekday, Figure 3.)
2. Which format sells most? (Format 2: its 17 hubs average 10,078 per open day; formats 1, 3, 4 average 6,816–6,919.) Are Sundays different? (Only 33 hubs ever open on Sunday.)
3. Do volume and app sessions move together? ($r = 0.825$ on open days.)

Our problem statements

P1 (primary). Within the same hub, is the promotion lift on Mondays different from the lift on Fridays?

P2. Is the 184-day data gap independent of hub format, i.e. is the missingness random?

P3. Is the distance to the nearest competitor associated with a hub's typical volume?

P4. When a competitor opens, does the hub's volume change relative to hubs with no new competitor over the same weeks?

How they are novel: the basic questions read one row or one column at a time. Ours need a contrast that no row records: a difference of lifts inside each hub (P1), the pattern of *missing* rows (P2), and a hub's change measured against control hubs over the same calendar weeks (P4). P3 aggregates two years to one value per hub.

Train vs test schedule (the test file has no outcome)

Rows	Open	Promo	School closed	Holiday
Train, all 900 days	82.9%	38.3%	17.4%	3.2%
Train, last 42 days	80.0%	35.7%	7.5%	6.4%
Train, 20 Jun–31 Jul 2014	85.6%	34.9%	27.6%	0.0%
Test, 20 Jun–31 Jul 2015	86.0%	35.7%	28.5%	0.0%

Two facts that shape the tests. (i) Days of one hub are dependent: the median within-hub correlation of $\log(1+V)$ is 0.65 at lag 1 and 0.55 at lag 14, but only 0.02 at lag 7, because promotions alternate weekly. (ii) The test window is the summer school break (school closure 28.5% vs 17.4% overall; 27.6% in the same weeks of 2014), so it is seasonal rather than a new regime.

5. Statistical inferences

Common to all tests. $y = \ln(1 + \text{OrderVolume})$ on open days with volume > 0. Because hub-days are dependent (Section 4), each test reduces the data to *one number per hub* (n hubs). We write $\bar{x}$ for a sample mean, s for a sample standard deviation (divisor $n-1$), SE for a standard error, μ, ρ, π for population values, and δ for the smallest difference that matters. Every statistic is $(\text{estimate} - \text{null value})/SE$, except χ^2. $\alpha = 0.05$. All tests are **two-sided**: an earlier version of these tests on raw volume had already shown the directions, so choosing a side now would be choosing it after seeing the data. We fixed the analysis choices in a committed plan (`analysis_plan_r1.md`) before computing these statistics. With four tests, Bonferroni uses $\alpha/4 = 0.0125$. Every statistic was computed with `scipy.stats` and by the formula shown; they agree to 4 significant figures. The δ values are our assumptions, with the reason given.

Test 1 (P1): is the promotion lift different on Monday and Friday?

a) Hypotheses, written before running the test

H_0	$\mu_d = 0$
H_1	$\mu_d \neq 0$
Significance level / side	$\alpha = 0.05$, two-sided
Test	paired t-test (one-sample t on d_i)
Smallest difference that matters	$\delta = 0.05$ on the log scale ($\approx 5\%$ of volume)

For hub i, $L_{i,\text{Mon}} = \bar{y}_{i,\text{Mon,promo}} - \bar{y}_{i,\text{Mon,no promo}}$ and likewise $L_{i,\text{Fri}}$, on open, non-holiday days outside reopening windows (each of the 4 cells ≥ 4 days); $d_i = L_{i,\text{Mon}} - L_{i,\text{Fri}}$ and μ_d is its population mean. e^d is the ratio of the Monday promotion multiplier ($e^{L_{i,\text{Mon}}}$) to the Friday one, so $e^d - 1$ is how much larger the Monday gain is. $\delta = 5\%$: below that, the choice of which weekday carries the promotion budget would not change.

b) Other tests we considered

The same hub appears in all four cells, so the data are paired; σ is unknown, so we use t.

Also considered	Why we did not use it
Welch t on hub-days	treats 262,170 dependent days (lag-1 $r = 0.65$) as independent; p would be far too small
Pooled two-sample t	the two weekdays come from the same hubs, so the groups are not independent
z-test	needs the population σ
Wilcoxon signed-rank, sign test	kept as robustness checks: $p = 6.0 \times 10^{-184}$; $d_i > 0$ for 1,112 of 1,115 hubs

c) Result

$$t = \frac{\bar{d} - 0}{s_d/\sqrt{n}} = \frac{0.2691}{0.09932/\sqrt{1115}} = \frac{0.2691}{0.002974} = 90.47.$$

Estimate	$\bar{d} = 0.269$: **Monday's promotion gain is 30.9% larger** (volume $\times 1.600$ on Monday, $\times 1.222$ on Friday; ratio $1.600/1.222 = 1.309$)
95% confidence interval	0.263 to 0.275 (30.1% to 31.6%)
Test statistic (d.f.)	$t = 90.47$ (1,114); $t_{0.975,1114} = 1.962$
p-value	$< 10^{-300}$
Decision	**reject** H_0 (also under Bonferroni)

The whole interval lies beyond $\delta = 0.05$, so the difference is large enough to matter. The normal Q-Q plot of d_i is close to a straight line (Figure 5), and with $n = 1{,}115$ the CLT makes $\bar{d}$ close to normal. **Replication:** 2013 only, $\bar{d} = 0.249$; 2014–15 only, 0.286; odd and even HubID, 0.267 and 0.272; with holidays and reopening days included, 0.270. All reject. On raw volume, $+2{,}169$ (95% CI $2{,}105$–$2{,}233$). **Placebo:** shuffling the promotion labels within hub $\times$ weekday 200 times gives a mean estimate of 0.0002; 7.0% of the 200 shuffles reject at 0.05, close to the 5% expected. **Shared calendar check:** the promotion calendar is the same for every hub, so hubs share date shocks. Taking the *week* as the unit (64 promotion and 52 non-promotion weeks, contrast = cross-hub Monday mean minus Friday mean): an F-test rejects equal variances ($F = 0.435$, d.f. 63 and 51, $p = 0.0018$), so we use Welch: $t = 10.6$ (d.f. 84.9), estimate 0.284, 95% CI 0.231–0.337, $p = 2.7 \times 10^{-17}$. The interval is 9 times wider, but it still lies wholly beyond δ.

Test 2 (P2): is the 2014-H2 data gap independent of hub format?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\pi_1 = \pi_2 = \pi_3 = \pi_4$ (gap independent of format)
$\boldsymbol{H_1}$	not all π_f are equal
Significance level / side	$\alpha = 0.05$; non-directional (upper tail of χ^2)
Test	χ^2 test of independence, 4×2 table, one row per hub ($n = 1{,}115$)
Smallest difference that matters	5 percentage points in a format's gap rate

π_f is the share of format-f hubs with no rows in 2014-H2 (overall 16.1%). $\delta = 5$ points, because a shift that size changes the format mix of the hubs observed in 2014-H2 enough to bias any format comparison over that period. Rule fixed in advance: the smallest expected count is 2.74 (< 5, format 2 has 17 hubs), so the p-value comes from 100,000 label permutations with the margins fixed; as a check, format 2 is merged into format 1.

b) Other tests we considered

Also considered	Why we did not use it
Two-proportion z	tests one contrast, not four groups; used below for the follow-up estimate
Asymptotic χ^2 only	an expected count below 5 weakens the approximation (we report it too)
Fisher's exact	standard form is 2×2; the permutation test is its Monte Carlo version for 4×2
χ^2 goodness of fit	compares one variable with fixed shares: the wrong question

c) Result

$$\chi^2 = \sum_{\text{cells}} \frac{(O - E)^2}{E} = 21.42,$$

where O and E are the observed and expected hub counts (E = row total $\times$ column total$/n$), with $(4-1)(2-1) = 3$ d.f. and critical value $\chi^2_{0.95,3} = 7.81$. Effect size: Cramér's $V = \sqrt{\chi^2/(n \cdot \min(4-1, 2-1))} = \sqrt{21.42/1115} = 0.139$. Gap rates are 14.0%, 5.9%, 9.5% and 23.3% for formats 1–4.

Estimate	format 4 vs rest: $\hat{\pi}_4 - \hat{\pi}_{\text{rest}} = 0.233 - 0.129 = 0.104$; Cramér's $V = 0.139$
95% confidence interval	0.053 to 0.154 (Wald, $SE = 0.0257$; Newcombe 0.055–0.156); V: 0.086–0.202
Test statistic (d.f.)	$\chi^2 = 21.42$ (3)
p-value	2.1×10^{-4} (permutation); 8.6×10^{-5} (asymptotic)
Decision	**reject** H_0 (also under Bonferroni)

The whole interval lies beyond 5 points, so the imbalance is large enough to matter. **Check and replication:** merged 3×2 table, $\chi^2(2) = 20.62$, $p < 10^{-4}$; odd and even HubID, $\chi^2(3) = 10.88$ ($p = 0.015$) and 11.07 ($p = 0.012$). **Implication:** the missingness is not random. Format 4 falls from 31.2% of all hubs to 28.6% of the hubs observed in 2014-H2, so any comparison over that period leans towards formats 1–3. This is why we never impute the gap, and why T4 re-

quires 30 observed days in each window.

Test 3 (P3): is competitor distance associated with typical volume?

a) Hypotheses, written before running the test

H_0	$\rho = 0$
H_1	$\rho \neq 0$
Significance level / side	$\alpha = 0.05$, two-sided
Test	t-test on Pearson's r, one row per hub ($n = 1{,}112$)
Smallest difference that matters	$\lvert\rho\rvert = 0.10$

ρ is the population correlation between $x_i = \ln(\text{distance})$ and $y_i =$ hub i's mean $\log(1+V)$. Three hubs with no distance are left out (complete case). Distance has skewness 2.93, so we use its log. $\delta = 0.10$, because an association explaining under 1% of the variation between hubs would not inform where to site a hub.

b) Other tests we considered

Also considered	Why we did not use it
Spearman ρ_s	robustness check (detects any monotone trend): $\rho_s = -0.037$, $p = 0.21$
Pearson on raw scale	a few far hubs dominate (skewness 2.93): $r = -0.045$, $p = 0.14$
Regression slope t	gives the identical t
Welch t, near vs far	a cut-off throws away information and is arbitrary

c) Result

$$t = \frac{(r-0)\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.08187\sqrt{1110}}{\sqrt{1-0.00670}} = -2.74.$$

Estimate	$r = -0.082$ (closer competitor, higher volume)
95% confidence interval	-0.140 to -0.023 (Fisher z)
Test statistic (d.f.)	$t = -2.74$ (1,110); $t_{0.975,1110} = 1.962$
p-value	0.0063
Decision	reject H_0 (also under Bonferroni, $0.0063 < 0.0125$)

The interval straddles -0.10, so we cannot tell whether the association is large enough to matter; at most it explains about 2% of the variation. Controlling for format leaves it unchanged (partial $r = -0.082$). It replicates: $r = -0.100$ on 2013 data and -0.067 on 2014–15 data. An earlier version of this test, Spearman on raw volume, failed to reject ($\rho_s = -0.032$, $p = 0.29$). The difference is explained by Figure 4 (a post-hoc look, not a test): the effect comes from the closest decile alone (122 hubs at distance ≤ 250, $+15.9\%$), while deciles 2–10 stay within $\pm 4.4\%$. With 5% of each tail trimmed, $r = -0.055$ ($p = 0.080$). A positive link between nearby competitors and demand is not competition helping: competitors cluster where shoppers already are (dense shopping streets), a hidden third factor.

Test 4 (P4): does a competitor opening change volume? (difference-in-differences)

a) Hypotheses, written before running the test

H_0	$\mu_{\text{DiD}} = 0$
H_1	$\mu_{\text{DiD}} \neq 0$
Significance level / side	$\alpha = 0.05$, two-sided
Test	paired t-test on DiD_i, $n = 144$ hubs
Smallest difference that matters	3% of volume (0.03 on the log scale)

Only the opening *month* is known, so that month is skipped. For a treated hub, $\Delta_i = \bar{y}_{\text{post}} - \bar{y}_{\text{pre}}$ over the 56 days before and the 56 days after the opening month (at least 30 observed days in each). C_i is the same change averaged over control hubs in the same calendar windows, and $\text{DiD}_i = \Delta_i - C_i$, so season, holidays and the shared promotion calendar cancel. Treated: openings from March 2013 to March 2015 with full windows, 144 of the 180 openings inside the data. Controls: 498 hubs whose competitor opened before 2012 or after June 2015 (418–498 per window). The 1900 sentinel and the 354 unknown dates are left out. $\delta = 3\%$, because a smaller fall is within the hub's ordinary week-to-week variation and would not call for a response.

b) Other tests we considered

Also considered	Why we did not use it
Before/after, no control (our earlier design)	season is confounded with "after": the control change alone varies across windows with SD 0.093, as large as the effect. It gives -5.6% (CI -7.8 to -3.4%)
Welch t, treated vs controls	ignores the pairing of each hub with its own calendar window
Wilcoxon signed-rank on DiD_i	robustness check (the Q-Q plot shows a heavy left tail): $p = 1.9 \times 10^{-17}$

c) Result

$$t = \frac{\overline{\text{DiD}} - 0}{s/\sqrt{n}} = \frac{-0.09218}{0.1170/\sqrt{144}} = \frac{-0.09218}{0.009752} = -9.45.$$

Estimate	$\overline{\text{DiD}} = -0.092$: **volume 8.8% lower after an opening**
95% confidence interval	-0.111 to -0.073 (-10.5% to -7.0%)
Test statistic (d.f.)	$t = -9.45$ (143); $t_{0.975,143} = 1.977$
p-value	9.0×10^{-17}
Decision	reject H_0 (also under Bonferroni)

The whole interval lies beyond -3%, so the fall is large enough to matter. **Sensitivity and replication:** 28-day windows -9.1%; 84-day windows -8.4%; same-format controls -8.8%; reopening days excluded -8.4%; odd and even HubID -9.0% and -8.6%; openings in 2013 and in 2014–15 -8.3% and -9.3%. All reject. **Placebo:** moving every opening 12 months earlier (92 hubs) gives -0.7% (CI -2.0 to $+0.6\%$), $t(91) = -1.08$, $p = 0.28$. We fail to reject, and the CI lies inside $\pm 3\%$: the design does not invent a fall where none should exist.



Figure 5: T1 differences are close to normal; T4 differences have a heavy left tail, so we also report Wilcoxon (same decision).

All four tests together

Test	Statistic (d.f.)	p	Estimate [95% CI]	Matters?
T1	$t = 90.47$ (1,114)	$< 10^{-300}$	$+30.9\%$ [30.1, 31.6] (ratio of multipliers)	yes
T2	$\chi^2 = 21.42$ (3)	2.1×10^{-4}	$+10.4$ points [5.3, 15.4]	yes
T3	$t = -2.74$ (1,110)	0.0063	$r = -0.082$ $[-0.140, -0.023]$	uncertain
T4	$t = -9.45$ (143)	9.0×10^{-17}	-8.8% $[-10.5, -7.0]$	yes

All four raw p-values are below the Bonferroni level 0.0125, and Holm gives the same four rejections. **T3 is the fragile one: it rests on one decile.** **What would change our mind:** T1, a week-level interval reaching below 0.05; T3, the closest-decile effect vanishing once a measure of shopper density is added; T4, placebo dates producing falls of similar size.

6. Limitations

- **Association, not cause.** The business sets one promotion calendar for all hubs, so promotion weeks may differ from other weeks in ways we cannot see. Competitors choose where to open, and T3 suggests they choose busy places. DiD assumes treated and control hubs would have moved in parallel; the placebo supports this but cannot prove it.
- **Shared dates.** All hubs share the calendar, so hub-level standard errors ignore cross-hub correlation. The week-level T1 check had a standard error 9 times larger. T4's control subtraction removes shared date shocks; T2 and T3 use one static value per hub.
- **Rows left out.** 166,269 closed and 54 open-zero rows; the non-random 2014-H2 gap (T2); 36 of the 180 openings (incomplete windows) and the 354 hubs with an unknown opening date (T4); 3 hubs without a distance (T3).
- **Month-precision event dates.** The opening month is skipped, so a fall that starts and ends within it would be missed.
- **Provenance and units.** The data is renamed public data, and the unit of `OrderVolume` is unstated. Our effects are relative (%), so they do not depend on the unit.
- **Choices made after EDA.** The questions came after exploration and an earlier raw-scale run, so all tests are two-sided and the plan is not a true pre-registration. The T3 decile breakdown is post hoc.
- **Scope.** January 2013 to June 2015 only; the test file's summer weeks are covered only through earlier summers.